A Mathematical Compendium of Algorithms, Machine Learning, Optimization, and Complexity

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Notation and Conventions. Let $\mathbb{R}$ denote the real numbers and $\mathbb{R}^n$ the n-dimensional Euclidean space. Vectors are denoted by boldface lowercase letters (e.g., $\mathbf{x}$), and matrices by boldface uppercase letters (e.g., $\mathbf{A}$). The transpose of a vector or matrix is denoted by $^\top$. The gradient of a scalar function f is ∇f, and its Hessian is $\mathbf{H}$ or $\nabla^2 f$. The indicator function of a set A is $\mathbb{I}(A)$. For asymptotic notation, n denotes the input size. The natural logarithm is log, and $\|\cdot\|_p$ denotes the L^p norm.

Part I
Complexity Theory and Algorithmic Analysis

In algorithmic analysis, we abstract away hardware specifics to study resource scaling (time and space) in terms of the input size n.

1 Asymptotic Notation (Formal Definitions)

Let $f, g : \mathbb{N} \to \mathbb{R}_{>0}$ be functions.

- **Big-O (O):** $f(n) = O(g(n))$ if $\exists c > 0, n_0 \in \mathbb{N}$ such that $\forall n \geq n_0$, $0 \leq f(n) \leq c \cdot g(n)$.
- **Big-Omega (Ω):** $f(n) = \Omega(g(n))$ if $\exists c > 0, n_0 \in \mathbb{N}$ such that $\forall n \geq n_0$, $0 \leq c \cdot g(n) \leq f(n)$.
- **Big-Theta (Θ):** $f(n) = \Theta(g(n))$ if $f(n) = O(g(n))$ and $f(n) = \Omega(g(n))$.
- **Little-o (o):** $f(n) = o(g(n))$ if $\lim_{n\to\infty} \frac{f(n)}{g(n)} = 0$.
- **Little-omega (ω):** $f(n) = \omega(g(n))$ if $\lim_{n\to\infty} \frac{f(n)}{g(n)} = \infty$.

2 The Master Theorem

For recurrences of the form $T(n) = aT(n/b) + f(n)$, where $a \geq 1, b > 1$, and $f(n)$ is asymptotically positive:

1. If $f(n) = O(n^{\log_b a - \epsilon})$ for some $\epsilon > 0$, then $T(n) = \Theta(n^{\log_b a})$.
2. If $f(n) = \Theta(n^{\log_b a})$, then $T(n) = \Theta(n^{\log_b a} \log n)$.
3. If $f(n) = \Omega(n^{\log_b a + \epsilon})$ for some $\epsilon > 0$, and if $af(n/b) \leq kf(n)$ for some constant $k < 1$ and all sufficiently large n, then $T(n) = \Theta(f(n))$.

3 Complexity Classes

- **P:** Problems solvable by a deterministic Turing machine in $O(n^k)$ time for some k.
- **NP:** Problems where a given solution can be verified in polynomial time by a deterministic Turing machine.
- **NP-Complete:** A problem is NP-Complete if it is in NP and every problem in NP is polynomial-time reducible to it. (Cook-Levin Theorem establishes SAT as the first NP-Complete problem).
- **NP-Hard:** Problems at least as hard as NP-Complete problems, but not necessarily in NP.
- **co-NP:** Problems where a "no" certificate can be verified in polynomial time.
- **PSPACE:** Problems solvable in polynomial space.
- **BPP (Bounded-Error Probabilistic Polynomial Time):** Problems solvable by a randomized algorithm with error probability $\leq 1/3$ in polynomial time.
- **#P (Sharp-P):** Counting problems associated with NP decision problems.

4 Algorithmic Paradigms

- **Randomized Algorithms:**
  - *Las Vegas:* Always correct, but runtime is a random variable.
  - *Monte Carlo:* Bounded runtime, but correctness is probabilistic.
- **Approximation Algorithms:** An algorithm is a ρ-approximation for an optimization problem if it runs in polynomial time and outputs a solution whose cost is within a factor ρ of the optimal cost.
- **Amortized Analysis:** Bounding the average time per operation over a worst-case sequence of operations.

Part II
Taylor Series and Mathematical Approximations

Taylor series are fundamental to numerical optimization, algorithmic analysis, and function approximations in ML.

5 The General Taylor Series

The Taylor series of a function $f(x)$ infinitely differentiable at a point a is:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

When $a = 0$, this is called the Maclaurin series.

6 The Multivariate Taylor Series

For a multivariate function $f : \mathbb{R}^n \to \mathbb{R}$, the second-order Taylor expansion around $\mathbf{x}_0$ is:

$$f(\mathbf{x}) \approx f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)^\top (\mathbf{x} - \mathbf{x}_0) + \frac{1}{2}(\mathbf{x} - \mathbf{x}_0)^\top \mathbf{H}(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0)$$

where $\nabla f(\mathbf{x}_0)$ is the gradient vector and $\mathbf{H}(\mathbf{x}_0)$ is the Hessian matrix. This is the basis for Newton's Method in optimization.

7 Essential Maclaurin Series

- $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ for all $x \in \mathbb{R}$.
- $\log(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}$ for $x \in (-1, 1]$.
- $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ for $|x| < 1$.
- $(1+x)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n$ for $|x| < 1$.
- $\sin(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$.
- $\cos(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$.
- **Stirling's Approximation:** $n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$, which implies $\log(n!) = n\log n - n + O(\log n)$.

Part III
Probability and Statistics

8 Core Definitions and Rules

- **Bayes' Theorem:** $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$.
- **Law of Total Probability:** $P(B) = \sum_i P(B|A_i)P(A_i)$.
- **Expectation:** $\mathbb{E}[X] = \sum_x xP(X = x)$ or $\int xf(x)dx$.
- **Variance:** $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.
- **Covariance:** $\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$.
- **Tower Property:** $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X|Y]]$.

9 Concentration Inequalities

- **Markov's Inequality:** For non-negative X and $a > 0$, $P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$.
- **Chebyshev's Inequality:** $P(|X - \mathbb{E}[X]| \geq a) \leq \frac{\text{Var}(X)}{a^2}$.
- **Chernoff Bounds:** For $X = \sum X_i$ with independent Bernoulli variables, $P(X \geq (1+\delta)\mu) \leq \exp\left(-\frac{\delta^2\mu}{2+\delta}\right)$.
- **Hoeffding's Inequality:** For independent random variables $X_i \in [a_i, b_i]$, $P(|\bar{X} - \mathbb{E}[\bar{X}]| \geq t) \leq 2\exp\left(-\frac{2n^2t^2}{\sum(b_i-a_i)^2}\right)$.
- **Jensen's Inequality:** If g is convex, $g(\mathbb{E}[X]) \leq \mathbb{E}[g(X)]$.

10 Key Distributions

- **Gaussian:** $\mathcal{N}(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$.
- **Multivariate Gaussian:** $\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{d/2}|\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$.
- **Exponential Family:** $p(x|\theta) = h(x)\exp(\eta(\theta)^\top T(x) - A(\theta))$.

11 Information Theory

- **Entropy:** $H(X) = -\sum P(x)\log P(x)$.
- **Cross-Entropy:** $H(P, Q) = -\sum P(x)\log Q(x)$.
- **KL Divergence:** $D_{KL}(P||Q) = \sum P(x)\log\frac{P(x)}{Q(x)}$, with the identity $H(P, Q) = H(P) + D_{KL}(P||Q)$.

Part IV
Combinatorics and Discrepancy Theory

12 Fundamental Counting

- **Permutations:** $P(n, k) = \frac{n!}{(n-k)!}$.
- **Combinations:** $\binom{n}{k} = \frac{n!}{k!(n-k)!}$.
- **Vandermonde's Identity:** $\binom{m+n}{r} = \sum_{k=0}^{r}\binom{m}{k}\binom{n}{r-k}$.
- **Stars and Bars:** $\binom{n+k-1}{k-1}$ ways to distribute n indistinguishable balls into k distinguishable bins.

13 The Probabilistic Method

- **Union Bound:** $P(\cup_i A_i) \leq \sum_i P(A_i)$.
- **Lovász Local Lemma (LLL):** Provides conditions under which the probability of avoiding a set of "bad events" with limited dependencies is strictly positive.
- **Method of Conditional Expectations:** A derandomization technique that fixes variables one by one to maintain the expected value.

14 Discrepancy Theory

- **Standard Definition:** $\text{disc}(\mathcal{F}) = \min_{\chi \in \{\pm 1\}^n} \max_{S \in \mathcal{F}} |\sum_{x \in S} \chi(x)|$.
- **Matrix Form:** Minimizing $\|A\mathbf{x}\|_\infty$, where A is the incidence matrix and $\mathbf{x} \in \{\pm 1\}^n$.
- **Spencer's Theorem:** For n elements and m sets, $\text{disc}(\mathcal{F}) = O(\sqrt{m \log(n/m)})$.

Part V
Optimization and Gradient Methods

15 Convexity

- **First-Order Condition:** $f(y) \geq f(x) + \nabla f(x)^\top (y - x)$.
- **Second-Order Condition:** The Hessian is positive semi-definite, $\mathbf{H} \succeq 0$.

16 Gradient Descent (GD)

- **Update Rule:** $\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \nabla \mathcal{L}(\mathbf{w}_t)$.
- **Convergence (Smooth Convex):** $O(1/\epsilon)$ iterations to reach ϵ-accuracy.
- **Stochastic Gradient Descent (SGD):** $\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \nabla \mathcal{L}_i(\mathbf{w}_t)$, where $\nabla \mathcal{L}_i$ is an unbiased estimate of the true gradient.

17 Newton's Method

- **Update Rule:** $\mathbf{w}_{t+1} = \mathbf{w}_t - [\mathbf{H}\mathcal{L}(\mathbf{w}_t)]^{-1} \nabla \mathcal{L}(\mathbf{w}_t)$.
- **Convergence:** Quadratic convergence near the minimum.

18 Regularization

- **L2 (Ridge):** $\Omega(\mathbf{w}) = \frac{\lambda}{2}\|\mathbf{w}\|_2^2$.
- **L1 (Lasso):** $\Omega(\mathbf{w}) = \lambda\|\mathbf{w}\|_1$, which promotes sparsity.
- **Lagrangian Formulation:** For $\min f(x)$ subject to $g_i(x) \leq 0$, the Lagrangian is $\mathcal{L}(x, \alpha) = f(x) + \sum \alpha_i g_i(x)$.
- **KKT Conditions:** Stationarity, Primal Feasibility, Dual Feasibility, and Complementary Slackness.

Part VI
Machine Learning Specifics

19 Bias-Variance Decomposition

$$\mathbb{E}[(y - \hat{f}(x))^2] = (\text{Bias}[\hat{f}(x)])^2 + \text{Var}[\hat{f}(x)] + \sigma^2_{\text{irreducible}}$$

- **Overfitting:** Characterized by high variance and low bias.
- **Underfitting:** Characterized by high bias and low variance.

20 Gradients of Common Loss Functions

- **MSE:** $\mathcal{L} = \frac{1}{2}(y - \hat{y})^2$; gradient with respect to $\hat{y}$ is $\hat{y} - y$.
- **Binary Cross-Entropy:** $\mathcal{L} = -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})]$; gradient with respect to logits is $\hat{y} - y$.
- **Softmax + Categorical Cross-Entropy:** Gradient with respect to logits is $\hat{\mathbf{y}} - \mathbf{y}$.

21 Matrix Calculus Identities

- $\frac{\partial}{\partial \mathbf{x}}(\mathbf{a}^\top \mathbf{x}) = \mathbf{a}$.
- $\frac{\partial}{\partial \mathbf{x}}(\mathbf{x}^\top \mathbf{A}\mathbf{x}) = (\mathbf{A} + \mathbf{A}^\top)\mathbf{x}$.
- $\frac{\partial}{\partial \mathbf{X}} \operatorname{Tr}(\mathbf{A}\mathbf{X}) = \mathbf{A}^\top$.
- $\frac{\partial}{\partial \mathbf{X}} \log|\det(\mathbf{X})| = (\mathbf{X}^{-1})^\top$.

22 Singular Value Decomposition (SVD)

For any matrix $\mathbf{A}$, the SVD is

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top.$$

This is the foundation of Principal Component Analysis (PCA) and low-rank approximation (Eckart-Young-Mirsky theorem).